Reliability-Oriented Proposal Refinement for Sparse Radar-Based Three-Dimensional Object Detection

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Abstract

Radar-only three-dimensional (3D) object detection is attractive for all-weather automotive perception, but sparse, sensor-dependent measurements make proposal selection unreliable. Filtering can remove useful boxes, classification confidence can disagree with localization quality, and average gains can hide dataset- or seed-specific regressions. We formulate proposal reliability in terms of proposal availability, confidence quality, and geometric consistency, and develop a controlled refinement framework around these dimensions. The framework preserves borderline candidates, recovers score-capped residual proposals, aligns confidence with localization quality, and restricts voting to supported primary boxes. We evaluate the method on Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar using three training seeds and AP_{R40} at a three-dimensional intersection-over-union threshold of 0.50. The strict reliability route improves the RDAR baseline in all 12 matched dataset-seed comparisons, yielding a mean gain of 0.9726 AP (bootstrap 95% confidence interval, [0.6468, 1.3536]). A separate high-performance route increases macro AP from 35.3552 to 39.1214, although one TruckScenes seed decreases. Progressive ablation, calibration diagnostics, point-dropout tests, threshold sensitivity, runtime profiling, and qualitative analysis link these outcomes

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to specific proposal mechanisms. These results show that strict reliability and maximum average accuracy are distinct objectives: constrained voting provides consistent paired gains at measurable post-processing cost, whereas confidence quality alignment delivers larger average gains and improved calibration without guaranteeing uniform improvement across diagnostics.

Keywords: Sparse radar 3D object detection, proposal preservation, proposal voting, calibration, robustness

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1. Introduction

Automotive perception must continue to function when illumination, weather, or viewing geometry weakens the visual evidence available to a camera. Millimeter-wave radar provides direct range and radial-velocity measurements and remains operational in several conditions that are difficult for optical sensors. Recent 4D radar datasets have consequently expanded radar perception from small proof-of-concept collections to diverse driving and cooperative settings [25, 29]. This progress makes radar-only 3D detection a useful research problem in its own right rather than only an auxiliary branch of camera–radar fusion.

Radar point clouds, however, do not behave like dense lidar scans. A vehicle may be represented by few spatially irregular returns, and the number, amplitude, and Doppler distribution of those returns vary with aspect, distance, occlusion, sensor configuration, and preprocessing. These differences are visible across Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar, which cover distinct platforms and acquisition regimes. A detector trained on such measurements must infer a full 3D extent from incomplete evidence. The difficulty is therefore not confined to feature extraction. Small changes in assignment, confidence, score filtering, or non-maximum suppression (NMS) can determine whether the limited evidence supporting an object survives to the final ranked list.

Efficient lidar-style detectors provide a practical starting point for this problem. PointPillars converts points into a bird's-eye-view (BEV) pseudo-image and applies a two-dimensional convolutional backbone [5]. Its regular representation, modest parameter count, and mature implementation make controlled multi-dataset experiments feasible. Yet a conventional single-stage head is commonly optimized with a binary classification target while evaluation rewards both correct class assignment and accurate 3D localization. Two positive anchors can therefore receive similar objectness supervision even when one produces a much better box. Once these predictions reach NMS, the higher classification score can cause the inferior box to suppress the better localized alternative.

The suppression stage introduces a second asymmetry. Standard hard NMS makes an irreversible decision: a proposal that is removed cannot benefit from later evidence. Soft-NMS showed in two-dimensional detection that preserving or decaying overlapping candidates can be preferable to deleting them immediately [18]. The issue is amplified for sparse radar, where several imperfect boxes may each encode part of the same weak object. Conversely, simply lowering the score threshold and retaining every box is not a complete solution. A larger proposal pool increases false positives and can degrade ranking unless the added candidates are scored conservatively and their geometric agreement is used explicitly.

Confidence quality is the third part of the problem. IoU-aware detectors learn or construct scores that better reflect localization accuracy [16]. Quality Focal Loss and VarifocalNet similarly integrate localization quality into dense-detector classification targets [14, 15]. These methods establish that a score is not merely a class probability; it controls ordering, suppression, and the precision–recall curve. Calibration adds a related but different requirement: a numerically high confidence should correspond to an appropriate empirical success frequency [36]. A route can improve AP by reordering boxes while worsening calibration, or reduce calibration error without improving every range or sparsity group. The two properties must therefore be measured separately.

Most detector comparisons compress these decisions into one mean AP value. That summary is necessary but insufficient for a reliability claim. A method can obtain a higher macro mean because of a large gain on one dataset while regressing on another. It can also benefit from one training initialization and fail under a second. Sparse radar exacerbates this risk because the four datasets considered here differ markedly in density and measurement statistics. We therefore distinguish two questions. First, what configuration maximizes average accuracy? Second, what configuration improves a fixed baseline in every paired dataset-seed comparison? These questions define different operating points and should not be answered by relabeling the same result.

Rather than enlarging the backbone, this paper treats proposal selection as a reliability problem. The framework combines candidate-preserving NMS (M1), score-capped residual recovery (M2), quality-aligned scoring (M3), and a strict route that gates and votes only among supported primary proposals. These components assign distinct roles to proposal availability, confidence quality, and geometric consistency while keeping their effects separately testable.

We report two operating routes. The *strict route* is evaluated against the

RDAR baseline using a positive AP difference in every matched dataset-seed cell. The *high-performance route* uses quality-aligned training and a relaxed anchor configuration to maximize macro AP, but it is not assigned a no-regression claim when any paired result is negative. The strict threshold was selected within the same four-dataset protocol used for the main strict comparison; therefore, the resulting evidence supports a bounded cross-dataset, cross-seed consistency claim rather than an independent held-out guarantee.

The study combines a formal four-dataset, three-seed comparison with a seed-2028 progressive analysis of the refinement path. Calibration error, range and sparsity diagnostics, deterministic point-dropout tests, voting threshold sensitivity, runtime profiling, and qualitative BEV cases examine the mechanisms and failure boundaries. The contributions are as follows:

1. **Reliability formulation.** We decompose sparse-radar proposal reliability into availability, quality, and consistency. This formulation connects candidate preservation, residual recovery, confidence–localization alignment, and local geometric consensus without changing the compact PointPillars-style backbone.

2. **Constrained proposal refinement.** We develop an asymmetric evidence path in which recovered residuals are score capped, positive classification targets combine objectness with online 3D IoU, and strict voting updates only supported primary centers and headings. These constraints allow weak evidence to assist a detection without letting it dominate ranking or expand box geometry.

3. **Paired reliability evaluation.** We operationalize strict reliability as positive improvement over RDAR in every matched dataset–seed cell and evaluate it across four radar datasets and three seeds. The selected strict route improves all 12 pairs, with a mean gain of 0.9726 AP and a minimum gain of 0.1718 AP.

4. **Separated operating objectives.** We distinguish the strict route from a high-performance route that raises macro AP by 3.7662 but contains one seed-level regression. Calibration, corruption, threshold, efficiency, and failure-boundary analyses show why maximum average AP and finite-sample no-regression evidence should not be treated as the same engineering claim.

2. Related Work

2.1. Point, voxel, and pillar representations

The representation chosen for an irregular point cloud determines both the available geometric information and the computational cost of detection. PointNet established a permutation-invariant architecture for unordered point sets, and PointNet++ introduced local hierarchical aggregation [1, 2]. These models avoid quantization but must repeatedly sample and group points. In sparse radar, direct point processing is conceptually attractive because every return is valuable, yet the limited and highly variable number of returns can make neighborhood construction unstable.

Voxel methods regularize the input before convolution. VoxelNet learns a feature for each occupied voxel, whereas SECOND exploits sparse convolution to avoid processing empty space [3, 4]. The resulting three-dimensional feature grids retain height structure but impose a memory cost that depends on resolution. Submanifold sparse convolutions further reduce unnecessary activation growth in sparse networks [20]. These approaches are strong general-purpose 3D detection foundations, but their greater architectural capacity can obscure whether an improvement originates in representation or in the final proposal decision path.

PointPillars collapses the vertical dimension within each pillar and scatters the encoded features to a BEV pseudo-image [5]. The design sacrifices some explicit height resolution in exchange for a simple 2D backbone and fast inference. It is especially suitable here because the proposal pipeline can be modified while the feature extractor remains fixed, allowing proposal decisions to be studied independently of representation capacity.

More elaborate detectors improve proposal features through a second stage. PointRCNN generates point-based proposals and pools features within them, whereas PV-RCNN combines sparse voxels with keypoint abstraction [6, 7]. CenterPoint instead treats object centers as keypoints and reduces reliance on anchors [9]. These methods improve representation and geometric reasoning, but final AP still depends on score ordering and duplicate removal. Our study fixes a compact pillar backbone and moves the analytical emphasis to those final proposal decisions.

2.2. Proposal generation, suppression, and voting

Two-stage object detection made the proposal set an explicit intermediate representation. Faster R-CNN generates regions before classification and box regression, and Cascade R-CNN progressively raises the localization quality of its stages [10, 17]. In point clouds, VoteNet uses learned Hough voting to aggregate point evidence around object centers [8]. Although these architectures differ from our single-stage radar setting, they share an important premise: several uncertain local observations can jointly yield a more stable object hypothesis.

NMS converts a dense set of hypotheses into a ranked detection list. A hard variant selects the current highest-scoring box and deletes neighbors whose overlap exceeds a threshold. This procedure is efficient, but its errors are irreversible. Soft-NMS retains overlapping boxes and decays their scores, demonstrating that immediate deletion is not always optimal [18]. Sparse R-CNN goes further by learning a fixed proposal set and iterative proposal interactions [19]. Both lines of work support treating proposal survival as a design decision rather than an unchangeable implementation detail.

Box voting represents a complementary strategy. Instead of selecting one hypothesis and discarding the rest, voting estimates a consensus geometry from nearby predictions. In three-dimensional scenes, this operation must respect the different roles of centers, dimensions, and periodic headings. Our study therefore treats voting as a conservative proposal-level refinement rather than as a second high-capacity detection network.

Proposal preservation alone can increase false positives, while voting alone cannot reconstruct a proposal that was previously removed. The current framework therefore links survival, recovery, scoring, and consensus in a defined sequence. M1 keeps the candidate pool open, M2 restores complementary evidence with capped scores, M3 orders proposals by localization-aware confidence, and the strict vote uses local agreement only after those earlier steps. The staged construction is a principal distinction from applying a single alternative NMS threshold.

2.3. Target assignment and localization-quality scoring

Dense detectors must decide which anchors or spatial locations supervise each object. FCOS removes explicit anchors, whereas ATSS adapts the positive-sample threshold to the statistics of candidate anchors [12, 13]. These studies show that target assignment is not a neutral preprocessing step: it determines which locations learn objectness and regression. In sparse radar, small perturbations of a predicted box can produce a large relative change in 3D IoU, making a rigid binary target particularly coarse.

Focal Loss addresses the imbalance between abundant easy negatives and scarce hard examples by modulating binary cross entropy [11]. It does not, by itself, distinguish a well-localized positive from a poorly localized positive. IoU-Net predicts localization quality as a separate quantity and uses it to improve ranking [16]. Generalized Focal Loss instead embeds a continuous quality value in the classification target [14]. VarifocalNet learns an IoU-aware classification score whose magnitude reflects both object presence and localization accuracy [15]. These approaches motivate the main trainable component of our framework.

These studies motivate the quality-aware component examined here. We use three-dimensional localization quality to reshape proposal ordering while retaining an objectness signal for assigned positives. The implementation and its training safeguards are defined in the Method section.

Quality-aware ranking is related to, but not synonymous with, calibration. Calibration evaluates whether confidence magnitude agrees with empirical correctness, while AP depends primarily on ordering. Modern neural networks can be accurate and still overconfident [36]. Object detection complicates the definition further because correctness depends on both class and localization threshold. For this reason, we report ECE as a diagnostic for the high-performance route and do not infer calibration from an AP increase. The strict voting route is evaluated for paired AP consistency, not promoted as a calibration technique.

2.4. Radar benchmarks and multimodal context

Automotive radar research now spans radar spectra, range–angle maps, point clouds, and dense radar tensors. Early learning systems demonstrated that radar-specific representations contain sufficient information for classification and detection [22, 21]. Range–angle processing retains a regular sensor-domain structure, whereas point-cloud conversion makes radar compatible with established 3D detection toolchains. The present study uses the latter representation to focus on proposal behavior across multiple public data sources.

K-Radar provides 4D radar measurements under varied weather conditions, and Enhanced K-Radar studies density reduction and accessibility [25, 26]. View-of-Delft and TJ4DRadSet extend the available road-user and driving scenarios [27, 28]. V2X-Radar introduces infrastructure, vehicle, and cooperative perspectives [29]. These datasets differ in coverage, density, sensing geometry, and labeling. Such variation is inconvenient for a single-dataset leaderboard but valuable for testing whether a proposal rule is tied to one measurement regime.

Multimodal approaches use radar as complementary evidence. RADIANT associates radar measurements with image features, while Radar Voxel Fusion integrates radar in a spatial 3D representation [30, 31]. PointPainting enriches point features with image semantics, and radar-aware painting adapts the strategy to automotive radar [32, 33]. Cross-modal augmentation and contrastive objectives can further stabilize representations [34, 35]. These methods address feature complementation and sensor fusion; our question is orthogonal. We deliberately test radar-only predictions so that an apparent proposal improvement cannot be attributed to semantic information imported from a camera.

The resulting gap is specific. Existing literature provides strong backbones, adaptive assignment, localization-aware scoring, suppression variants, and radar datasets, but few studies evaluate their proposal-level interaction under a paired multi-seed, multi-dataset reliability criterion. We address this gap with a controlled PointPillars-style backbone, the RDAR proposal stream as a fixed reference, and separate accuracy-maximizing and no-regression routes.

3. Method

3.1. Problem formulation

Let a radar frame be represented by a point set $X = \{\mathbf{x}_n\}_{n=1}^{N_x}$, where each point contains spatial coordinates and the radar attributes supplied by the corresponding dataset conversion. The detector maps X to a set

$$\mathcal{P} = \{(\mathbf{b}_i, s_i, c_i)\}_{i=1}^{N_p}, \tag{1}$$

where $\mathbf{b}_i = (x_i, y_i, z_i, l_i, w_i, h_i, \theta_i) \in \mathbb{R}^7$ is a 3D oriented box, $s_i \in [0, 1]$ is its confidence, and c_i is its class. The experiments consider the car class and use the dataset-specific training and validation partitions frozen in the project protocol.

Conventional training minimizes classification, regression, and direction losses, whereas inference additionally applies score filtering and NMS. Our objective is not simply to increase the number of raw candidates. We seek a ranked set that satisfies three properties: plausible weak candidates remain available for recovery; confidence preferentially ranks better-localized boxes; and redundant hypotheses can provide geometric consensus rather than only suppressing one another. Reliability is then assessed by paired AP differences across datasets and seeds.

We distinguish *proposal availability*, *proposal quality*, and *proposal consistency*. Availability concerns whether a candidate reaches the final processing stages. Quality concerns whether its score agrees with its localization. Consistency concerns whether neighboring predictions support a common geometry. M1 and M2 primarily address availability, M3 addresses quality, and strict consistency refinement acts on the final proposal set. The division is functional: M1, M2, and the final gate/vote are proposal-level operations, while M3 changes training.

3.2. Baseline detector and framework overview

The common detector follows PointPillars [5]. Points are grouped into vertical pillars with a voxel size of $[0.25, 0.25, 5.0]$ m within the baseline range

$[0, -25, -3, 60, 25, 2]$ m. At most 32 points are retained per pillar, with 16,000 training voxels and 40,000 test voxels. A PillarVFE maps each occupied pillar to 64 channels, PointPillarScatter places the features on the BEV grid, and a 2D backbone with block depths $[3, 5, 5]$, strides $[2, 2, 2]$, and channels $[64, 128, 256]$ produces the detection features. Three upsampling branches return 128 channels each and are concatenated before the anchor head.

The baseline head predicts classification, seven box residuals, and a two-bin direction label. Its loss weights are 1.0 for classification, 2.0 for localization, and 0.2 for direction. For the frozen Astyx configuration, the car anchor is approximately $[4.003, 1.800, 1.510]$ m with bottom height -0.180 m, and the positive/negative matching thresholds are 0.60/0.45. Dataset-specific configuration files provide the corresponding data paths and anchor statistics, while the network family and evaluation rule remain fixed.

The complete processing path has three progressive construction modules and one strict consistency stage. M1 changes post-processing so borderline proposals survive. M2 combines the output with a residual stream. M3 trains a quality-aligned confidence. The strict route then reweights primary proposals against an expert stream and performs local box voting. A separate high-performance route uses the M3 head with relaxed positive assignment and a K-Radar prior configuration. It is an alternative operating point, not another name for the strict route.

This distinction resolves an otherwise ambiguous module diagram. The seed-2028 progressive analysis ends in the high-performance branch because it documents the accuracy-oriented construction path. The formal strict route starts from RDAR-compatible proposals and applies consistency refinement, because its selection objective is 12-of-12 positive paired differences. Neither analysis is used to imply that the two endpoints are identical.

The complete decision path is summarized in Fig. 1 and should be read from left to right. M1 and M2 preserve and recover proposal evidence, whereas M3 changes the confidence target used by the accuracy-oriented branch. The lower branch is a distinct strict endpoint that applies expert-consistency gating and conservative local voting to RDAR-compatible primary proposals. The branch point is important because the strict route is not obtained by appending voting to the final high-performance configuration.

3.3. M1: candidate-preserving NMS

The baseline configuration uses a score threshold of 0.10 and a very strict NMS IoU threshold of 0.01. Such a setting produces a compact output, but it can delete

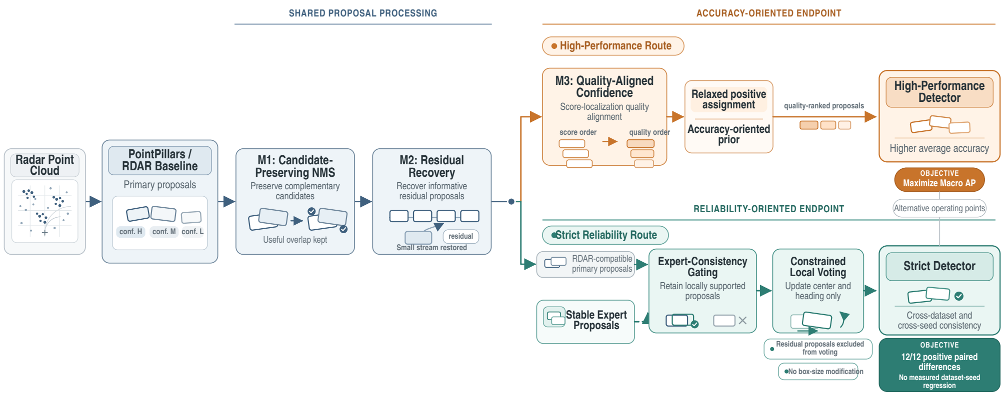

Figure 1: Overview of the reliability-oriented proposal-refinement framework. Shared candidate preservation and residual recovery lead to distinct accuracy-oriented and reliability-oriented endpoints. The strict route operates on RDAR-compatible primary proposals rather than extending the high-performance route.

useful evidence before any complementary proposal is considered. M1 lowers the score threshold to zero and increases the NMS threshold to 0.50. The pre-NMS and post-NMS limits remain 4096 and 500, respectively. Thus M1 changes which boxes remain available, not the maximum reported proposal count.

Let π order boxes by score. Hard NMS accepts the highest remaining box and removes a box j whenever

$$\mathrm{IoU}_{3D}(\mathbf{b}_{\pi(i)}, \mathbf{b}_j) > \tau_{\mathrm{nms}}. \tag{2}$$

Increasing τ_{nms} from 0.01 to 0.50 delays this deletion and retains moderately overlapping alternatives. This resembles the preservation motivation of Soft-NMS [18], although our implementation does not use continuous score decay. The retained alternatives become inputs to recovery, quality-aware ranking, and voting.

Removing the initial score threshold does not mean every proposal is promoted to a high-confidence detection. The post-NMS limit remains finite, subsequent ranking uses the learned score, and residual additions in M2 are explicitly capped below the positive primary-score range. M1 should therefore be read as a candidate-access operation. It creates the opportunity for later modules to use weak boxes; it does not assert that weak boxes are correct.

3.4. M2: residual proposal recovery

M1 cannot restore a hypothesis that exists only in a complementary prediction stream. M2 takes a primary set $\mathcal{P} = \{(\mathbf{b}_i, s_i)\}$ and a residual set $\mathcal{R} = \{(\mathbf{r}_j, t_j)\}$. Frame identifiers and ordering are validated before fusion. The residual boxes are sorted by t_j, and only the top $K_r = 50$ are processed.

For each residual proposal, we find

$$i^\star(j) = \arg\max_i \mathrm{IoU}_{3D}(\mathbf{r}_j, \mathbf{b}_i). \tag{3}$$

When the best IoU is at least $\tau_r = 0.10$, the residual and primary geometries are combined using normalized score weights. For the six non-angular box parameters,

$$\widetilde{\mathbf{b}}_{j,1:6} = \frac{s_{i^\star}\mathbf{b}_{i^\star,1:6} t_j \mathbf{r}_{j,1:6}}{s_{i^\star} + t_j}. \tag{4}$$

Heading is π-periodic for an oriented rectangular box. We therefore use doubled-angle circular averaging:

$$\widetilde{\theta}_j = \frac{1}{2}\,\mathrm{atan2}\left(\sum_{k\in\{i^\star,j\}} w_k \sin 2\theta_k, \sum_{k\in\{i^\star,j\}} w_k \cos 2\theta_k\right). \tag{5}$$

This avoids cancellation when equivalent headings fall on opposite sides of the numerical angle boundary. If the best overlap is below 0.10, the residual geometry is appended unchanged.

Recovered proposals are deliberately prevented from dominating the primary ranking. Let $s^+_{\min}$ be the smallest positive primary score in the evaluated prediction set and $t_{\max}$ the maximum residual score in the frame. The assigned recovery score is

$$s^{\mathrm{rec}}_j = \left(\frac{s^+_{\min}}{2}\right)\left(0.5 + 0.5\frac{t_j}{\max(t_{\max}, \epsilon)}\right), \tag{6}$$

where adjacency denotes multiplication and $\epsilon = 10^{-12}$ prevents division by zero. The factor in parentheses lies in $[0.5, 1]$, so every residual score is at most half the smallest positive primary score.

3.5. M3: localization-quality-aligned scoring

Binary positive labels do not distinguish anchors by localization accuracy. M3 replaces the positive classification target with an online 3D-IoU target. For each positive anchor, the predicted residual and assigned regression target are decoded into $\mathbf{b}_i$ and $\mathbf{g}_i$. Their aligned quality is

$$q_i = \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{g}_i) \in [0, 1]. \tag{7}$$

The IoU is detached from the classification graph, so the classification loss does not back-propagate through the IoU operation into box regression.

Directly using q_i can make a valid positive behave nearly like a negative while the regressor is immature. We retain a residual objectness component:

$$\widehat{q_i} = \rho + (1 - \rho) q_i^{\gamma}, \qquad \rho = 0.55, \quad \gamma = 1.0. \tag{8}$$

Negatives receive zero and ignored anchors are excluded. The parameter ρ controls the minimum target for an assigned positive. At $\rho = 1$, the target reduces to binary objectness; at $\rho = 0$, it is pure IoU quality. The selected value retains both signals.

Let z_i be the classification logit and $p_i = \sigma(z_i)$. The quality focal term is

$$\mathcal{L}_{\mathrm{QF}} = \frac{1}{N_+} \sum_{i \in C} \left| \widehat{q_i} - p_i \right|^{\beta} \mathrm{BCEWithLogits}(z_i, \widehat{q_i}), \qquad \beta = 2, \tag{9}$$

where the juxtaposition after the modulation factor denotes multiplication, C is the set of cared anchors, and N_+ is the number of positives, clipped below by one. This construction follows the general quality-aware principle of Generalized Focal Loss [14], but uses online aligned 3D IoU and an explicit objectness floor for sparse radar.

The full training loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{QF}} + 2\mathcal{L}_{\mathrm{box}} + 0.2\mathcal{L}_{\mathrm{dir}}. \tag{10}$$

M3 does not add a second inference branch. The original classification logit is trained to express the combined objectness–quality target, leaving the 4.830 M parameter count unchanged. The quality-aligned route retains the baseline assignment configuration and uses the candidate-preserving post-processing settings. The separate high-performance route combines M3 with 0.50/0.35 positive/negative assignment thresholds and the accuracy-oriented prior configuration.

The role of M3 is ranking. A well-localized proposal should receive a larger target than a poorly localized proposal assigned to the same class. Better ranking can improve which proposal survives NMS and where a detection appears on the precision–recall curve. It does not guarantee that confidence is calibrated in every sense, so ECE is measured independently in Section 5.5.

3.6. Strict consistency refinement

The strict route applies two proposal-level operations to a primary RDAR set: an expert-consistency quality gate and local box voting. The expert is a stable BEV-gated detector with the same output parameterization. Its purpose is not to replace the primary detector but to provide a second assessment of whether a proposal is spatially supported. The strict route is selected by the paired no-regression gate described below, not by the performance of a single seed.

3.6.1. Expert-consistency quality gate

Let $\mathcal{E} = \{(\mathbf{e}_j, u_j)\}$ be the expert predictions. The final $K_r = 50$ entries of the combined primary set are reserved for residual additions and are not gated. For every other primary proposal, the best expert match is

$$j^{\star}(i) = \arg\max_j \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{e}_j). \tag{11}$$

If the maximum IoU v_i is at least $\tau_e = 0.30$, the score becomes

$$s_i' = s_i^{1-\alpha} u_{j^\star}^{\alpha} v_i^{\eta}, \qquad \alpha = 0.30, \quad \eta = 0.25. \tag{12}$$

This geometric product requires agreement among primary confidence, expert confidence, and overlap. The small IoU exponent avoids allowing modest geometric noise to erase otherwise supported radar proposals. If $v_i < 0.30$, the proposal is retained but down-weighted:

$$s_i' = \kappa s_i, \qquad \kappa = 0.50. \tag{13}$$

Keeping unmatched proposals instead of deleting them preserves the conservative character of the pipeline. The gate affects ranking but does not move a box.

3.6.2. Local proposal voting

The voting stage refines geometry using the gated scores. For each primary box i, a neighborhood is defined by

$$\mathcal{N}_i = \{j : \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{b}_j) \geq \tau_v\}, \qquad \tau_v = 0.24. \tag{14}$$

If $|\mathcal{N}_i| \leq 1$, the box is unchanged. Otherwise, $w_j = \max(s_j', 10^{-8})$ and

$$\mathbf{c}_{i,1:6} = \frac{\sum_{j\in\mathcal{N}_i} w_j \mathbf{b}_{j,1:6}}{\sum_{j\in\mathcal{N}_i} w_j} \tag{15}$$

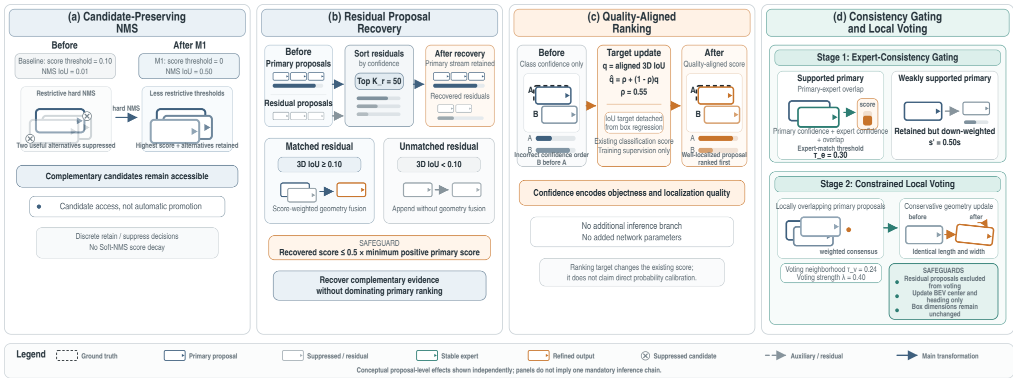

Figure 2: Proposal-level effects and safeguards of M1, M2, M3, and strict consistency refinement. The panels illustrate candidate preservation, score-capped residual recovery, localization-quality-aware ranking, and expert-supported center-and-heading voting without box resizing.

defines the consensus geometry. Consensus heading is again calculated in doubled-angle space. The strict xy mode updates only the BEV center:

$$\mathbf{b}'_{i,xy} = (1-\lambda)\mathbf{b}_{i,xy} + \lambda\mathbf{c}_{i,xy}, \qquad \lambda = 0.40. \tag{16}$$

Heading is interpolated circularly with weights $1-\lambda$ and λ. Height, center z, length, width, and height remain unchanged. Restricting the update prevents the consensus from artificially shrinking or expanding boxes to satisfy the evaluation threshold.

The IoU neighborhood threshold is lower than the 0.50 evaluation threshold. This is intentional: multiple radar hypotheses for the same vehicle can have modest 3D overlap because of height and dimension noise while still agreeing in BEV. Threshold sensitivity is nevertheless evaluated over a closed grid rather than justified only after observing the selected setting. The final 50 residual proposals remain outside the voting pool, ensuring that consistency refinement cannot undo M2 by moving or suppressing the recovered tail.

The proposal-level effects of the four operations are illustrated in Fig. 2. The reader should compare candidate survival after M1, the score-capped residuals introduced by M2, the confidence reordering produced by M3, and the restricted center-and-heading update used by strict voting. The illustration also marks two safeguards: residual proposals cannot enter the voting pool, and voting does not resize a primary box.

3.7. Operating routes and selection criteria

The *RDAR baseline* denotes M1 followed by M2 under the frozen recovery protocol. The *quality-aligned route* applies M3 and provides the intermediate trainable comparison. The *high-performance route* uses the same quality-focal formulation with relaxed positive assignment and the accuracy-oriented prior configuration. It is chosen by macro AP across the four datasets. Its intended use is to show the performance available when average accuracy is prioritized.

The *strict route* applies the expert gate and local vote. Let $A^m_{d,r}$ be AP for method m, dataset d, and seed r. A candidate is strict only when

$$\Delta_{d,r} = A^m_{d,r} - A^{\mathrm{RDAR}}_{d,r} > 0 \quad \forall (d, r) \in \mathcal{D} \times \mathcal{S}, \tag{17}$$

where $|\mathcal{D}| = 4$ and $|\mathcal{S}| = 3$. This definition contains no tolerance for a small negative difference. It is stronger than requiring a positive macro mean, but remains bounded to the tested data, splits, and seeds.

Selection by Eq. (17) is not a claim that the strict route has the highest AP. Indeed, the high-performance route has a larger macro mean. The two routes answer different engineering questions: whether a configuration offers the largest average gain and whether it avoids a regression in the available paired tests.

The exact operations enabled by each reported configuration are listed in Table 1. This table should be consulted when comparing the quality-aligned, high-performance, and strict endpoints because they optimize different objectives and are not successive aliases for one model. In particular, relaxed positive assignment belongs to the high-performance route, whereas expert gating and local voting define the strict route.

3.8. Algorithmic summary

For clarity, the complete procedure is summarized below.

Step 1: Encode the radar frame with PillarVFE, scatter pillar features to BEV, and predict anchor classification, box residuals, and direction.

Step 2: During M3 training, decode positive boxes, compute aligned 3D IoU, form the soft target in Eq. (8), and optimize the combined loss. The baseline route retains binary classification.

Step 3: Apply candidate-preserving NMS with zero initial score threshold, $\tau_{\mathrm{nms}} = 0.50$, and a 500-box post-NMS limit.

Table 1: Definitions and enabled operations of the configurations used in the progressive analysis and formal route comparison. A check mark denotes that the corresponding operation or setting is active.

Configuration	M1	M2	M3	Relaxed assignment	Accuracy prior	Expert gate	Vote	Role in the study
PointPillars screen	–	–	–	–	–	–	–	Construction starting point
M1	✓	–	–	–	–	–	–	Candidate-preservation screen
RDAR	✓	✓	–	–	–	–	–	Frozen internal baseline
Quality-aligned	✓	✓	✓	–	–	–	–	Trainable quality comparison
High-performance	✓	✓	✓	✓	✓	–	–	Maximum macro-AP endpoint
Strict	✓	✓	–	–	–	✓	✓	12-of-12 paired-positive endpoint

M1: candidate-preserving NMS; M2: residual proposal recovery; M3: localization-quality-aligned scoring. Relaxed assignment uses car positive/negative thresholds of 0.50/0.35. The accuracy prior denotes the route-specific prior configuration used by the high-performance endpoint. The strict endpoint starts from RDAR-compatible proposals and therefore does not include M3 or the relaxed-assignment/high-performance prior changes.

Step 4: Retain the top 50 complementary residual proposals. Fuse a residual box with its best primary match when 3D IoU is at least 0.10; otherwise append it. Assign the conservative score in Eq. (6).

Step 5: For the strict route, match each non-residual primary proposal to the expert stream, apply Eq. (12), and down-weight unmatched boxes by 0.50.

Step 6: Form local neighborhoods at $\tau_v = 0.24$ and update BEV center and heading using Eq. (16). Return the residual tail unchanged.

Step 7: Rank the resulting predictions and evaluate them with the identical AP_{R40} implementation used for every compared method.

3.9. Training and implementation details

Training uses the OpenPCDet codebase and an Adam one-cycle optimizer for 160 epochs. The frozen baseline configuration uses a per-GPU batch size of four, initial learning rate 0.003, weight decay 0.01, momentum 0.9, one-cycle momenta $[0.95, 0.85]$, warm-up fraction 0.4, division factor 10, and gradient norm clipping at 10. Random world flips, rotations in $[-\pi/4, \pi/4]$, and scaling in $[0.95, 1.05]$ are applied during training. Ground-truth sampling is disabled in the frozen radar configuration used for the controlled comparisons.

All models are trained independently with seeds 2026, 2027, and 2028. The formal comparison uses the same data partitions, class definition, metric, checkpoint epoch, and evaluation code for each paired method. Proposal-level scripts validate frame order before combining prediction files. This check is important because an apparently valid fusion between misaligned frame lists could silently produce meaningless boxes.

The progressive construction study is intentionally narrower. It holds the seed at 2028 and adds M1, M2, quality alignment, and the high-performance calibration variant sequentially. It is used to identify the contribution scale of the stages, not to establish that every intermediate module is positive for all seeds. The formal strict result instead evaluates the final strict route on all 12 paired cells.

Computational cost is reported in three parts. Trainable parameters distinguish model capacity. A fixed profiler measures detector forward execution including OpenPCDet post-processing but excluding AP computation and result serialization. A separate timer measures only the strict voting loop and excludes prediction-file input/output. Keeping these measurements separate prevents a fast detector result from hiding an expensive post-processing stage.

3.10. Design invariants and safeguards

The implementation enforces several invariants that are easy to omit from a high-level pipeline diagram. First, each prediction set remains frame aligned. All proposal-combination scripts compare the complete sequence of frame identifiers and stop when it differs. A mismatch is not repaired by truncation or sorting because either operation could pair predictions from different scenes.

Second, primary and residual proposals have different ranking privileges. Primary scores are learned by the detector. Residual scores are bounded by Eq. (6) and appended as the final 50 entries. The quality gate and voting stage calculate $N_{\text{primary}} = N_{\text{total}} - 50$ and operate only on the primary prefix. This convention prevents a residual candidate from becoming both a source and a target of repeated refinement. It also makes the reported proposal counts reproducible.

Third, angle operations respect the geometry of a 3D box. For car detection, headings θ and $\theta + \pi$ represent the same undirected rectangular axis before the direction classifier resolves orientation. Linear averaging near the wrap boundary can return an orthogonal or unstable direction. Doubled-angle sine and cosine averages provide a continuous mean on this π-periodic space. The same representation is used for residual fusion and proposal voting.

Fourth, the strict vote is deliberately non-expansive in box dimensions. Only x, y, and heading are updated; z, l, w, and h are copied from the original proposal. This restriction limits the ways in which a vote can improve IoU. A more flexible all-parameter vote could fit the local cluster more closely but would also create a stronger risk of dataset-specific size bias. The current operation tests the narrower hypothesis that local center and orientation jitter are correctable.

Fifth, all thresholds are global within a route. The main formal comparison does not choose a different vote IoU or strength for each dataset. The closed sensitivity grid includes dataset-level outcomes, but one common setting is reported across Astyx, TruckScenes, V2X-Radar-V, and K-Radar. Dataset-specific compensation screens are analyzed separately and are not silently folded into the strict route.

These safeguards make the method less flexible than an unconstrained ensemble, but that is intentional. The contribution is a controlled proposal-refinement study. A method that changes the data order, score privileges, geometry, or thresholds independently for each result would be difficult to audit and would weaken the paired reliability claim.

4. Experimental Setup

4.1. Datasets and protocol

We evaluate radar-only car detection on four datasets. Astyx provides an automotive radar point-cloud benchmark [23]. MAN TruckScenes covers autonomous trucking scenes and a sensor geometry distinct from passenger-car platforms [24]. V2X-Radar-V contributes the vehicle-side subset of a cooperative 4D radar collection, and K-Radar contains 4D radar measurements across varied weather conditions [25, 29]. The frozen conversions are denoted Astyx, TruckScenes-mini, V2X-Radar-V-400, and K-Radar-400 in the experiment records. We use the shorter dataset names in the paper. Exact conversion names, configuration files, and frozen outputs are retained in the accompanying experiment package.

The scope of the cross-dataset comparison is defined in Table 2. The reader should use it to verify the frozen data conversion labels, sensor settings, radar inputs, spatial evaluation range, and three training seeds. The acquisition setting differs across datasets, whereas the car-class mapping, AP_{R40} metric, 0.50 3D-IoU match threshold, and paired-seed comparison rule remain common.

The primary metric is car AP_{R40} at 3D IoU 0.50. Forty-point interpolation reduces the coarseness of the older 11-point protocol, while the fixed 0.50 IoU threshold makes the proposal-localization requirement explicit. All methods use the same evaluation implementation and class mapping. AP values are reported on a 0–100 scale, so a difference of 1.0 denotes one AP point rather than one percentage relative to the baseline.

The formal experiment crosses four datasets with seeds 2026, 2027, and 2028. For each dataset, models are trained independently under the same optimization schedule and evaluated with the same class mapping and metric implementation. We report the mean and sample standard deviation over the three seeds. More

Table 2: Dataset scope and common protocol for the formal four-dataset, three-seed evaluation.

Dataset	Frozen conversion	Platform or acquisition setting	Model input	Spatial range $[x, y, z]$ (m)	Formal seeds
Astyx	Astyx	Automotive passenger-vehicle scenes	Radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
MAN TruckScenes	TruckScenes-mini	Autonomous trucking scenes	Radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
V2X-Radar-V	V2X-Radar-V-400	Vehicle-side cooperative-perception subset	4D radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
K-Radar	K-Radar-400	Driving scenes under varied weather conditions	4D radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028

All rows use the paper's frozen car-class mapping and are evaluated with AP_{R40} at a 3D-IoU match threshold of 0.50. Each method is trained independently for the three listed seeds, and strict reliability is calculated from matched method–RDAR differences within the same dataset and seed. The spatial range is the common model input range recorded in the frozen configurations, not a claim about the physical maximum range of each sensor.

importantly, strict reliability is assessed from paired differences between methods trained with the same seed. This pairing reduces some between-seed variation and exposes regressions that could be hidden by an average.

The evidence is ordered to limit circular interpretation. The formal three-seed comparison supports the principal claims. The seed-2028 progressive study explains the construction path but is not used as evidence of three-seed monotonicity. Calibration and recall-bin analyses diagnose scoring behavior, whereas point dropout and threshold grids provide stress and sensitivity tests. Runtime profiling and qualitative examples address deployment cost and mechanism. Secondary analyses are not promoted to formal claims solely because their outcomes are favorable.

4.2. Metric and statistical summaries

For a ranked list at a fixed 3D-IoU match threshold, precision and recall after the first k predictions are

$$\mathrm{Prec}(k) = \frac{\mathrm{TP}(k)}{\mathrm{TP}(k) + \mathrm{FP}(k)}, \qquad \mathrm{Rec}(k) = \frac{\mathrm{TP}(k)}{\mathrm{TP}(k) + \mathrm{FN}(k)}. \tag{18}$$

AP_{R40} samples the interpolated precision envelope at 40 recall positions. Because confidence determines the traversal order, M3 can change AP even when the set of decoded boxes is similar. Because voting changes geometry, the strict route can also change which predictions cross the 0.50 IoU matching threshold.

Table 3: Main three-seed comparison of RDAR and the quality-aligned, high-performance, and strict routes.

Route	Astyx	TruckScenes	V2X-Radar-V	K-Radar	Macro	ΔMacro	Strict 12/12
RDAR	32.8347 ± 1.4689	16.3671 ± 1.7480	41.7029 ± 1.1490	50.5163 ± 2.0546	35.3552	–	Reference
Quality-aligned	35.8438 ± 1.0680	16.9661 ± 1.2785	**44.4122 ± 2.5011**	57.6910 ± 2.5304	38.7283	+3.3731	No
High-performance	**36.9801 ± 0.9748**	16.8063 ± 1.1764	44.3786 ± 0.5122	**58.3208 ± 1.6943**	**39.1214**	**+3.7662**	No
Strict	33.6898 ± 1.7215	**17.2878 ± 1.7127**	42.1245 ± 1.0852	52.2091 ± 2.5654	36.3278	+0.9726	Yes

Values are AP_{R40} mean ± sample standard deviation over seeds 2026, 2027, and 2028 at a 3D-IoU threshold of 0.50. Macro is the unweighted mean of the four dataset means, and ΔMacro is measured from RDAR. Bold values mark the largest result in each dataset or aggregate column. "Strict 12/12" requires a positive paired difference from RDAR in every dataset–seed cell. The quality-aligned and high-performance tags correspond to `qflr55` and `q55rpa50_kprior`, respectively.

The macro result for method m is the unweighted average of the four dataset-level seed means:

$$\overline{A}^{m}_{\text{macro}} = \frac{1}{4}\sum_{d\in\mathcal{D}}\left(\frac{1}{3}\sum_{r\in\mathcal{S}} A^{m}_{d,r}\right). \tag{19}$$

This choice gives each dataset equal influence regardless of the number of frames or objects. It avoids allowing the largest validation partition to dominate, but it does not imply equal operational importance.

Standard deviations describe variation across the three training seeds. They are not confidence intervals on dataset performance. For the strict comparison, we additionally report the 12 paired differences, their mean and median, a bootstrap interval for the mean, and an exact sign test. The bootstrap interval is therefore based on a finite set of 12 paired cells, not on independent observations from an unlimited population. Effect sizes and the minimum paired difference remain primary; the p-value is not used to convert a small improvement into a practically large one. Because the strict threshold was selected within the same four-dataset protocol used for the formal comparison, the result is interpreted as bounded consistency evidence rather than an independent held-out guarantee.

5. Results

5.1. Main comparison

The formal three-seed results are summarized in Table 3. Dataset means and standard deviations quantify average performance and between-seed variation, while the unweighted macro column gives each dataset equal influence. The high-performance route provides the largest macro gain, whereas the strict route is the only reported endpoint selected by the paired no-regression criterion.

The formal three-seed comparison shows two distinct outcomes. RDAR obtains a macro mean of 35.3552. Strict proposal voting increases it to 36.3278, a gain of 0.9726 AP. The gain is modest relative to the high-performance branch, but every dataset-level mean increases: 0.8551 on Astyx, 0.9207 on TruckScenes, 0.4216 on V2X-Radar-V, and 1.6928 on K-Radar. TruckScenes is the only dataset on which the strict route also provides the best mean among the reported routes.

The quality-aligned route reaches 38.7283 macro AP. It provides large mean improvements on Astyx, V2X-Radar-V, and K-Radar, but the qflr55 configuration has a negative TruckScenes difference at seed 2027. It therefore supports the effectiveness of quality-aware ranking but not Eq. (17).

The high-performance route reaches 39.1214 macro AP, 3.7662 above RDAR. Its largest mean gain is 7.8045 AP on K-Radar, followed by 4.1454 on Astyx and 2.6758 on V2X-Radar-V. TruckScenes improves by only 0.4392 on average. At seed 2027, TruckScenes decreases from 18.3845 for RDAR to 17.1075, a difference of -1.2770. This single cell is sufficient to reject the strict label even though the macro result is highest.

Aggregate means do not show whether an operating route regresses under an individual initialization. Fig. 3 therefore displays the AP difference from RDAR for each of the 12 matched dataset–seed cells. All strict-route markers lie on the positive side of zero, whereas the high-performance route contains one negative TruckScenes result at seed 2027. The figure should be used to assess consistency; exact values and paired statistics are reported in Table 4.

The comparison illustrates why the two routes are both useful. If a deployment can accept dataset-specific validation and prioritizes average AP, the high-performance route is preferable. If the experimental objective is to avoid a measured regression across all available paired conditions, the strict route provides the relevant evidence. A single ranking of methods would erase this difference in engineering objective.

5.2. Dataset-level interpretation

Astyx shows the clearest response to candidate preservation and quality alignment: M1 adds 3.8383 AP and M3 adds a further 2.6778 AP in the seed-2028 construction. TruckScenes is the limiting dataset for the high-performance route because its seed-2027 result decreases, whereas strict voting remains positive in all three paired comparisons. V2X-Radar-V has the smallest strict mean gain, although quality alignment improves its mean by 2.7093 AP. K-Radar provides the largest headroom, contributing 7.8045 AP to the high-performance macro gain and

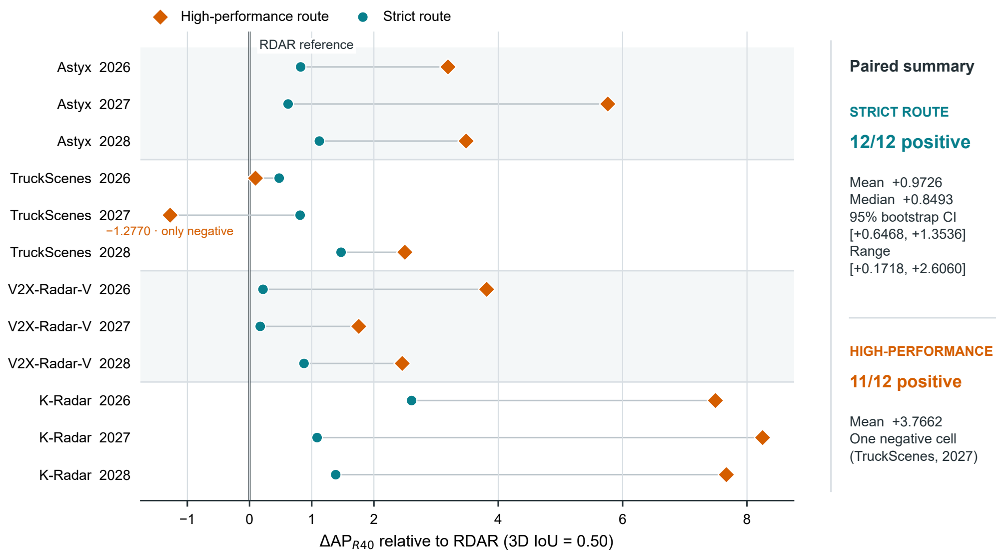

Figure 3: Matched dataset–seed AP differences for the strict and high-performance routes relative to RDAR. The strict route improves all 12 paired cells, whereas the high-performance route has a larger mean gain but one negative TruckScenes result at seed 2027. The bootstrap interval summarizes the mean of the 12 finite strict-route paired differences.

1.6928 AP to the strict-route gain. These dataset-specific patterns indicate that no single proposal mechanism dominates across all radar measurement regimes.

5.3. Paired reliability across datasets and seeds

The strict-route mean is expanded into all 12 matched comparisons in Table 4. The sign of every row, rather than only the average difference, determines whether Eq. (17) is satisfied. The table also reports the mean, median, minimum difference, bootstrap interval, and exact sign test so that effect magnitude and finite-sample consistency remain distinguishable.

The paired comparison expands the strict-route result. All 12 differences are positive. The smallest is 0.1718 on V2X-Radar-V seed 2027, and the largest is 2.6060 on K-Radar seed 2026. The mean over paired cells is 0.9726 and the median is 0.8493. Thus the conclusion does not depend on one extreme K-Radar improvement: removing the maximum cell would still leave every remaining difference positive.

We computed a nonparametric bootstrap interval by resampling the 12 paired differences and recalculating their mean. The 95% interval is [0.6468, 1.3536],

Table 4: Complete matched-seed reliability audit for the strict route.

Dataset	Seed	RDAR AP	Strict AP	ΔAP
Astyx	2026	32.7281	33.5503	+0.8222
Astyx	2027	31.4220	32.0422	+0.6202
Astyx	2028	34.3540	35.4768	+1.1228
TruckScenes	2026	15.4127	15.8892	+0.4765
TruckScenes	2027	18.3845	19.1980	+0.8135
TruckScenes	2028	15.3041	16.7762	+1.4721
V2X-Radar-V	2026	40.7802	40.9969	+0.2167
V2X-Radar-V	2027	42.9899	43.1617	**+0.1718**
V2X-Radar-V	2028	41.3385	42.2148	+0.8763
K-Radar	2026	51.3450	53.9510	**+2.6060**
K-Radar	2027	48.1767	49.2631	+1.0864
K-Radar	2028	52.0271	53.4132	+1.3861

Strict AP is the RDAR value plus the frozen matched difference for `m3rob_q15p25_viou0p24_s0p40`. Across the 12 cells, the mean, median, minimum, and maximum ΔAP are 0.9726, 0.8493, 0.1718, and 2.6060. The nonparametric bootstrap 95% interval for the mean is [0.6468, 1.3536]. The exact sign test gives one-sided $p = 0.000244$ and two-sided $p = 0.000488$; these p-values are auxiliary because the four datasets are not random population draws.

which remains above zero. An exact sign test of 12 positive differences gives a one-sided $p = 0.000244$ and a two-sided $p = 0.000488$ under an equal-probability sign null. These statistics are auxiliary because the four datasets are not random draws from a well-defined population and 12 cells do not justify a universal claim. Their role is to summarize the consistency of the finite paired experiment, not to replace effect sizes or dataset-level reporting.

V2X-Radar-V defines the narrowest margin. Its mean improvement is only 0.4216, and two seeds are below 0.22 AP. The strict conclusion is consequently sensitive to reproducibility of these small differences. We retain four decimal places in the audit trail, use identical evaluation code, and avoid rounding the individual cells before testing the gate. In a deployment decision, the magnitude should be considered together with the direction: the route is consistently positive under this protocol, not uniformly large.

5.4. Progressive module ablation

The seed-2028 construction path is traced in Fig. 4. The reader should compare both the macro trajectory and the four dataset-specific trajectories: M1 contributes most visibly on Astyx, M2 produces a small incremental change, and M3 supplies the largest macro step, driven especially by K-Radar. Because only one seed is shown, the figure explains module behavior but does not establish three-seed

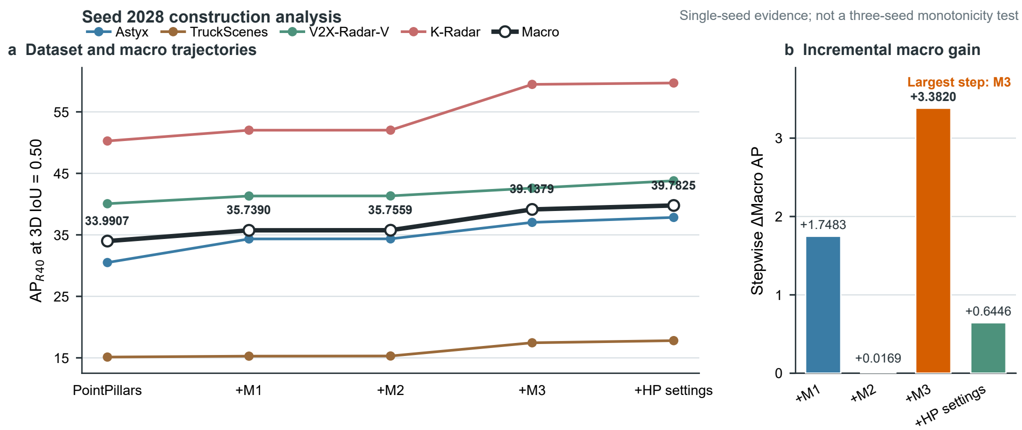

Figure 4: Progressive seed-2028 construction analysis across the four radar datasets and their macro average. Panel (a) traces AP through the PointPillars screen, M1–M3, and the final high-performance settings; bold labels mark the macro trajectory. Panel (b) reports stepwise macro changes relative to the immediately preceding configuration. This single-seed analysis explains module behavior but does not establish three-seed monotonicity.

monotonicity.

Exact values underlying Fig. 4 are given in Table 5. Incremental changes should be read relative to the immediately preceding configuration, not only relative to the starting screen. This exposes the small contribution of M2 and the much larger M3 step without promoting the single-seed analysis to a formal multi-seed claim.

The seed-2028 progressive study identifies the relative roles of the modules. M1 raises macro AP by 1.7483, with the largest dataset gain on Astyx. M2 adds only 0.0169 macro AP, so recovery should be interpreted as a conservative interface for complementary evidence rather than the main performance source.

M3 supplies the largest step, from 35.7559 to 39.1379 macro AP. Its gains are 2.6778 on Astyx, 2.1489 on TruckScenes, 1.2594 on V2X-Radar-V, and 7.4416 on K-Radar. The broad positive direction supports the intended quality-ranking mechanism, while the large K-Radar contribution reinforces the need for paired multi-seed analysis.

The final accuracy-oriented adjustment adds 0.6446 AP in this seed-2028 construction. Because TruckScenes seed 2027 is negative in the formal comparison, the progressive analysis is treated as a mechanistic screen rather than a factorial ablation or proof of cross-seed monotonicity.

Table 5: Numerical results for the seed-2028 progressive construction analysis.

(a) Dataset AP

Progressive setting	Astyx	Truck	V2X	K-Radar
PointPillars screen	30.5009	15.1291	40.0670	50.2659
+M1: candidate-preserving NMS	34.3392	15.2784	41.3191	52.0193
+M2: RDAR recovery	34.3540	15.3041	41.3385	52.0271
+M3: quality alignment	37.0318	17.4530	42.5979	59.4687
+High-performance settings	**37.8373**	**17.8028**	**43.7939**	**59.6960**

(b) Macro progression

Progressive setting	Macro	Stepwise Δ
PointPillars screen	33.9907	–
+M1: candidate-preserving NMS	35.7390	+1.7483
+M2: RDAR recovery	35.7559	+0.0169
+M3: quality alignment	39.1379	+3.3820
+High-performance settings	**39.7825**	+0.6446

All rows use seed 2028. Stepwise ΔMacro is measured from the immediately preceding row. The final row combines the M3 quality target with relaxed positive assignment and the accuracy-oriented prior configuration. This table documents the construction path and must not be interpreted as a three-seed factorial or as proof that every intermediate step is positive for all seeds.

5.5. *Calibration and diagnostic behavior*

The auxiliary diagnostics are collected in Fig. 5 but answer different questions. Panel (a) compares ECE and shows that the two quality-aware routes improve confidence calibration relative to RDAR, whereas strict voting is not supported by the same calibration claim. Panels (b)–(d) are addressed in the subsequent robustness and sensitivity subsections and should not be interpreted as additional samples in the formal 12-cell comparison.

ECE partitions predictions by confidence and compares mean confidence with empirical correctness in each bin [36]. The diagnostics show lower ECE for both quality-aware routes than for RDAR on all four seed-2028 datasets. The high-performance route improves ECE by 0.0749 on Astyx, 0.0407 on TruckScenes, 0.0354 on V2X-Radar-V, and 0.0845 on K-Radar. Across the three-seed diagnostic audit it produces 12 wins and no losses relative to RDAR, with a mean ECE reduction of 0.0504.

The quality-aligned route has the best V2X-Radar-V value, whereas the high-performance route is best on the other three datasets. Thus the final accuracy adjustment is not uniformly best calibrated even though it has the largest macro AP. This result reinforces the separation between ranking, calibration, and aggregate

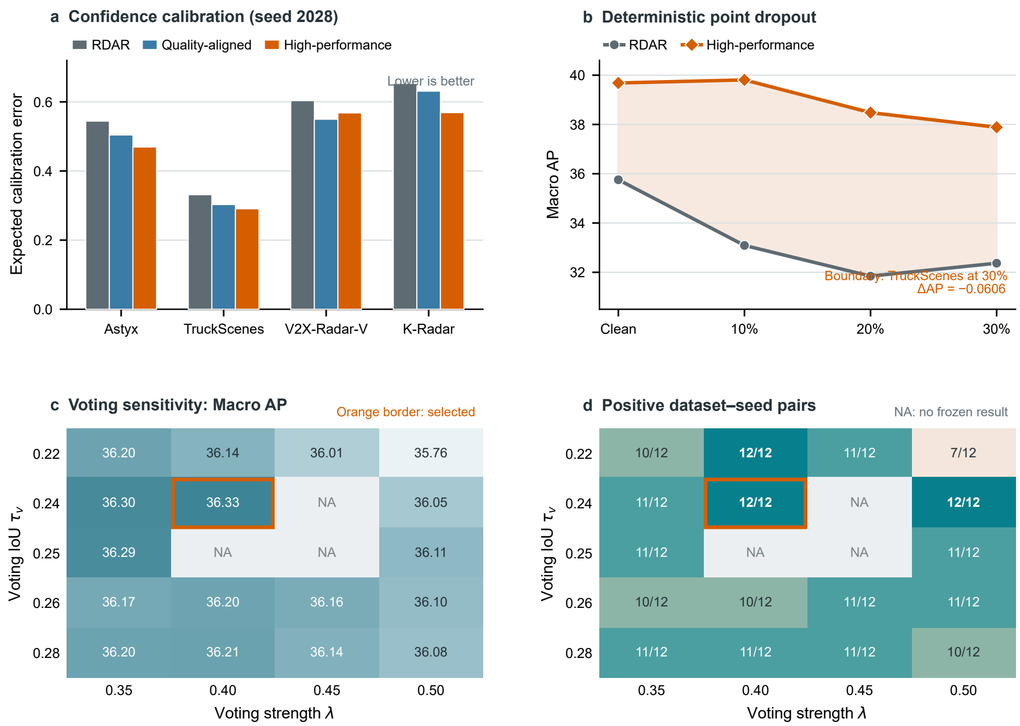

Figure 5: Diagnostic and boundary analyses. (a) Seed-2028 ECE for RDAR and the two quality-aware routes; lower values indicate better calibration. (b) Macro AP under deterministic inference-time point dropout. The frozen dropout comparison uses the available RDAR seed-2028 checkpoints and the high-performance seed-2027 checkpoints for Astyx, TruckScenes, and V2X-Radar-V and seed 2028 for K-Radar. (c,d) Macro AP and the number of positive dataset–seed pairs over the voting IoU–strength grid. Orange borders mark the selected $(\tau_v, \lambda) = (0.24, 0.40)$ setting; NA denotes a grid combination without a frozen result.

accuracy.

Strict proposal voting behaves differently. It wins all formal AP pairs but worsens ECE in all 12 diagnostic comparisons, with a mean unfavorable change of 0.0270. Voting shifts geometry and reweights proposal support after the confidence has been learned; it was not optimized to preserve probability calibration. We therefore make no calibration claim for the strict route.

Range and sparsity results are also mixed. Relative to RDAR, strict voting has 26 wins, 13 ties, and 9 losses across the audited range bins, and 18 wins, 19 ties, 8 losses, and 3 missing cells across sparsity bins. The high-performance route has 17 range-bin wins, 10 ties, and 21 losses, despite its strong AP and ECE. These counts show that aggregate ranking gains cannot be translated into a statement that every distant or sparsely observed target is recalled more often. The bin analysis is used as a failure boundary, not selectively filtered into a positive presentation.

5.6. Robustness and corruption tests

The macro response to deterministic point dropout is shown in Fig. 5(b). The high-performance route retains an average advantage at each tested dropout level, but the reader should note the single TruckScenes failure at 30% dropout. The panel therefore supports average-positive corruption behavior rather than a no-regression claim.

We simulated measurement loss by dropping radar points before voxelization with a fixed dropout seed of 3407. Models were not retrained or adapted to the corruption. The high-performance route retains a macro advantage at dropout rates of 10%, 20%, and 30%. Relative to its own clean checkpoint, its macro changes are +0.1242, −1.2030, and −1.7970 AP, compared with −2.6651, −3.9114, and −3.3868 for RDAR. The mean degradation advantage is 2.3624 AP.

At the dataset-by-dropout level, the high-performance route exceeds RDAR in 11 of 12 cells. The exception is TruckScenes at 30%, where RDAR obtains 12.0161 and the high-performance route obtains 11.9555, a difference of −0.0606. The stress test is therefore average-positive but not strict. In addition, the available dropout comparison uses the latest frozen checkpoints: seed 2028 for all RDAR models, seed 2027 for three high-performance datasets, and seed 2028 for K-Radar. It is supportive evidence rather than a paired-seed substitute for the formal paired comparison.

An exploratory vote applied directly to the dropout predictions repairs the single negative cell and yields 12 positive comparisons, but its smallest margin is only 0.0010 AP and it reduces the original high-performance result in 8 of 12 cells. Because the setting was examined in response to the failed cell, treating it as confirmatory robustness evidence would be post hoc. We exclude it from the main claim.

A separate closed compensation grid evaluates 192 combinations over range power, support scale, and fusion strength. The best settings differ: $(0, 8, 0.05)$ for Astyx, $(0, 4, 0.30)$ for TruckScenes, $(2, 1, 0.05)$ for V2X-Radar-V, and $(2, 8, 0.20)$ for K-Radar. The lack of one shared optimum is itself informative. Radar statistics are dataset dependent, and a physical-support correction selected on one sensor regime should not be assumed optimal on another.

We also screened which radar evidence most usefully re-ranks proposals. The best Astyx configuration combines all available physical signals and reaches

35.7324 AP in its screen. TruckScenes is best with return count alone at 18.3579. V2X-Radar-V prefers count and Doppler at 43.4016, whereas K-Radar prefers count and radar cross section at 54.2241. These screens are not directly comparable to the formal three-seed comparison because they use a separate screening protocol. Their value is diagnostic: a fixed semantic notion of "radar quality" does not translate to one optimal evidence combination on all datasets.

5.7. Voting-threshold sensitivity

The response surface in Fig. 5(c) shows how macro AP changes with voting IoU and vote strength, while Fig. 5(d) marks how many of the 12 paired cells remain positive. The selected setting lies inside a region containing more than one 12-of-12 configuration; stronger voting can increase the macro result while sacrificing the strict criterion.

The strict route could be fragile if its 12-of-12 result occurred only at one precise threshold. We therefore evaluated a closed grid over voting IoU and strength. Among the completed settings, $(\tau_v, \lambda) = (0.22, 0.40)$ reaches 36.1364 macro AP and is positive in all 12 cells; $(0.24, 0.50)$ reaches 36.0479 and is also 12-of-12 positive. The selected $(0.24, 0.40)$ strict configuration obtains the formal no-regression result. Neighboring strengths of 0.35 and several larger IoU thresholds produce only 10 or 11 positive cells.

This response surface supports two conclusions. First, all-pair positivity is not unique to one numerical point, because more than one nearby setting satisfies the gate. Second, the gate is not automatic: a slightly different strength can trade a higher macro value for one regression. We therefore report both the selected parameters and the grid boundary rather than claiming threshold insensitivity.

5.8. Qualitative cases

The mechanism-oriented BEV examples are organized in Fig. 6. Each row compares the RDAR and strict-route matched boxes against the same ground-truth target. The reader should inspect the relative center and heading changes rather than interpret the examples as an estimate of effect frequency. All three targets are deliberately borderline RDAR misses: their initial IoUs lie below 0.50, while the restricted strict-voting update moves them across the matching threshold.

The qualitative cases connect the voting operation to visible proposal changes. In Astyx frame 000499, the RDAR box has IoU 0.4203 and lies below the 0.50 matching threshold. Strict voting moves the BEV center toward the local consensus and raises IoU to 0.5235. TruckScenes frame 000267 shows a larger change, from

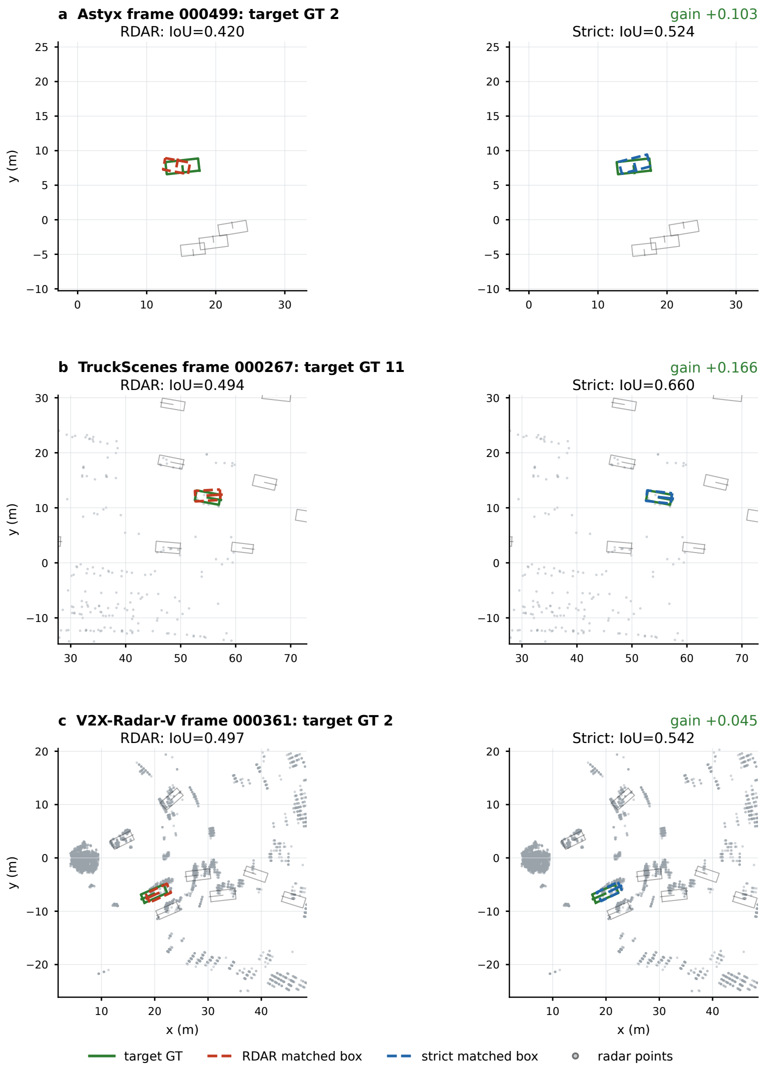

Figure 6: Representative selected BEV cases in which strict proposal voting converts an RDAR miss into a match. Each row compares the RDAR matched box (red dashed) and strict-route matched box (blue dashed) with the same ground-truth box (green): (a) Astyx frame 000499, (b) TruckScenes frame 000267, and (c) V2X-Radar-V frame 000361. Gray marks are radar points where the frozen artifact contains point measurements. These qualitative cases illustrate the center-and-heading refinement mechanism and are not estimates of its frequency.

0.4940 to 0.6599. V2X-Radar-V frame 000361 changes from 0.4970 to 0.5420. In each case a borderline localization becomes a valid match under the metric.

Table 6: Accuracy and computational cost under the fixed profiling protocol.

(a) Trainable parameter count

Configuration	Parameters	Parameter interpretation
RDAR detector	4.830 M	Primary detector and frozen internal baseline
Quality-aligned detector	4.830 M	M3 changes supervision, not network width
High-performance detector	4.830 M	Separate trained accuracy-oriented endpoint
Stable expert detector	4.868 M	Auxiliary detector used by the strict route
Strict voting operation	+0	Proposal-level operation with no trainable weights

(b) Separately profiled time per frame (ms)

Profiled stage	Astyx	TruckScenes	V2X-Radar-V	K-Radar
RDAR detector forward	4.740 ± 0.297	3.851 ± 0.311	4.834 ± 1.092	5.887 ± 2.409
High-performance detector forward	8.263 ± 1.163	7.880 ± 1.257	8.124 ± 1.334	8.601 ± 1.471
Strict voting loop	22.7095 ± 0.1031	22.0086 ± 0.1826	22.5393 ± 0.0471	22.8485 ± 0.1735

Hardware: NVIDIA GeForce RTX 3090. Detector-forward profiling uses batch size 4, two workers, three warm-up batches, and at most 20 measured batches; it includes OpenPCDet post-processing but excludes AP evaluation and result serialization. RDAR uses seed-2028 checkpoints; the high-performance profile uses seed 2027 for Astyx, TruckScenes, and V2X-Radar-V and seed 2028 for K-Radar. Voting is timed separately over 100 Astyx frames and 80 frames for each other dataset and excludes expert inference, prediction-file I/O, and evaluation. The independently profiled rows must not be summed and reported as a measured end-to-end strict-route latency.

The cases are selected illustrations, not independently sampled estimates of effect frequency. They do not prove that voting improves every hard object or every range group. Their proper role is mechanistic: they show that the restricted xy and heading update can change matching status without resizing a box, which is consistent with the formal paired AP result.

5.9. Efficiency

Accuracy and computational cost are compared in Table 6 under the same profiling protocol. The quality-aware routes retain the RDAR parameter count because M3 changes supervision rather than network width, whereas the strict route adds no trainable voting parameters but incurs measurable proposal-refinement latency. Separate timing columns prevent "parameter-free" from being misread as "zero-cost."

The quality-aligned and high-performance detectors preserve the 4.830 M parameter count of RDAR because the quality target changes supervision rather than adding a branch. The stable expert used by the strict route has 4.868 M parameters. The voting operation itself contains no trainable parameter.

In the fixed forward profiler, the high-performance detector remains below 9 ms per frame on all four datasets but is consistently slower than RDAR. The difference cannot be attributed to parameter count; it reflects the configuration and proposal processing performed during the measured forward path.

Strict voting adds about 22–23 ms per frame in the current implementation. The timer processes 500 primary boxes per frame using GPU 3D-IoU construction followed by a Python loop over proposals. Across the profiled subsets, it votes 47,790 of 50,000 primary boxes on Astyx, 37,277 of 40,000 on TruckScenes, 38,009 of 40,000 on V2X-Radar-V, and 38,457 of 40,000 on K-Radar. The large number of local neighborhoods explains why a mathematically simple operation can dominate runtime.

These data prevent two misleading claims. The strict route is not "zero-cost" merely because voting adds no weights, and detector latency cannot be combined with a separately timed vote to create a favorable number. An optimized batched or sparse-neighborhood kernel could reduce the overhead, but that optimization is future work. Under the present implementation, the strict route exchanges latency for paired reliability, while the high-performance route offers the more practical accuracy-oriented operating point.

6. Discussion

The central result is not that one post-processing setting dominates every alternative. It is that proposal handling accounts for a measurable and reproducible part of sparse-radar detector behavior, and that different proposal objectives lead to different operating points. Strict voting improves RDAR in every paired dataset-seed comparison, whereas quality-aligned training produces the larger macro gain. The distinction would be invisible if the paper reported only the best macro AP.

6.1. How the stages contribute

The progressive experiment suggests a hierarchical mechanism. Candidate preservation has a large initial effect because it changes which evidence is available. A later module cannot recover a box that no longer exists unless a second stream reproduces it. Relaxing the filter consequently increases AP on all four seed-2028 datasets before any trainable quality change is introduced. The magnitude varies by dataset, which is expected when baseline score and overlap distributions differ.

Residual recovery contributes little incremental AP in the progressive analysis. This does not support describing M2 as a major accuracy innovation. Its value is architectural and conservative: it formalizes how complementary boxes enter the output, how matched geometry is fused, and how scores are kept below the primary stream. Removing that control would make it difficult to distinguish recovery from simply appending enough low-quality predictions. The current evidence supports M2 as a stabilizing bridge rather than the dominant module.

Quality alignment accounts for the largest improvement. The mechanism is consistent with the loss design: binary labels encode presence, whereas the soft target differentiates positives by decoded 3D IoU. Because confidence controls ranking and NMS, better score–quality agreement can improve AP without changing the backbone or parameter count. The ECE results offer independent support, particularly the 12-of-12 diagnostic improvement of the high-performance route. They do not prove that every individual score becomes well calibrated, but they make a purely incidental ranking explanation less likely.

Strict voting addresses a different error. Its selected cases show proposals with nearly correct geometry that remain below the evaluation threshold. Consensus moves the BEV center and heading while preserving dimensions. This limited update can convert a near miss into a true positive without allowing a local cluster to invent a new box size. The paired gains are largest on K-Radar and smallest on V2X-Radar-V, indicating that the prevalence or correctability of localization jitter varies by measurement regime.

6.2. Reliability versus maximum accuracy

The high-performance route is more accurate on average and would be the natural choice if macro AP were the only objective. Calling it robust without qualification would nevertheless be misleading. TruckScenes seed 2027 regresses by 1.2770 AP even though the TruckScenes three-seed mean improves. The example demonstrates a general problem in model comparison: averaging over initializations can turn one substantial regression into a small positive mean.

The strict criterion avoids that presentation error by requiring every paired difference to be positive. It does not guarantee future no-regression. Datasets, seeds, training schedules, and thresholds form a finite test set, and a new condition could still fail. The term "strict" is therefore defined operationally by Eq. (17), not used as a synonym for universally safe. This bounded definition makes the claim reproducible and falsifiable.

The criterion also has a statistical limitation. The 12 cells combine four datasets and three seeds, but cells sharing a dataset are not independent in the same way as samples from unrelated populations. The sign test and bootstrap interval summarize the observed differences; they do not license a population-level causal conclusion about every radar sensor. More datasets, more seeds, and preregistered operating thresholds would strengthen the reliability assessment.

6.3. *Calibration and failure boundaries*

AP, ECE, recall, and localization diagnostics answer different questions. The high-performance route improves AP and ECE but loses more range-bin comparisons than it wins. Strict voting improves paired AP while worsening ECE. Neither result is contradictory. AP rewards ordering under a localization threshold; ECE measures the magnitude of confidence relative to correctness; binned recall measures whether targets are retrieved under a particular partition. A method can move the right boxes upward in the ranking while shifting score magnitudes in an unfavorable way.

This distinction changes how the method should be deployed. A downstream planner that treats confidence as a calibrated probability may prefer the quality-aligned branch or apply a separate calibration procedure. A pipeline that needs consistent AP under the tested conditions may prefer strict voting, provided its confidence values are not interpreted probabilistically without additional validation. The paper therefore avoids merging "quality alignment", "calibration", and "strict reliability" into one marketing claim.

Range and sparsity bins reveal another boundary. Sparse radar difficulty is often described as if fewer points or longer range should define a single monotonic axis. In practice, return count interacts with RCS, Doppler, aspect, occlusion, and the dataset conversion. The mixed bins and dataset-specific compensation optima indicate that one global physical correction is unlikely to cover all regimes. This finding favors sensor-aware validation over universal threshold transfer.

6.4. *Robustness interpretation*

Point dropout evaluates sensitivity to missing input measurements, but it is a synthetic perturbation. Randomly deleting points does not reproduce all physical causes of radar degradation: multipath can add returns, occlusion can remove them nonuniformly, and calibration error can shift their coordinates. The favorable macro degradation of the high-performance route therefore supports tolerance to one controlled sparsification process, not general adverse-weather invariance.

The single TruckScenes failure at 30% dropout is small in absolute AP but important in interpretation. Repairing it with a post-hoc voting setting would produce an attractive 12-of-12 result, yet the same vote harms eight other cells. We retain the negative result because it exposes the actual boundary: the high-performance route is robust on average under the test, not uniformly. This decision also keeps the confirmatory strict claim isolated from exploratory robustness tuning.

6.5. Efficiency and engineering use

The accuracy-oriented quality head does not increase model parameters, but its fixed-profile latency is approximately two to four milliseconds higher than RDAR depending on the dataset. It remains below nine milliseconds per frame on the profiled RTX 3090. The strict route has a more substantial cost. Its expert and voting stages require a second source of proposals and an additional 22–23 ms per frame for the current vote.

The overhead is an implementation property as well as a method property. The current script computes a dense primary-box IoU matrix and iterates over proposals in Python. A batched GPU kernel, spatial hashing, or pre-clustering could reduce the cost without changing Eqs. (12) and (16). Until such an implementation is measured, however, the strict route should be positioned for reliability-sensitive offline evaluation or systems whose latency budget can accommodate the refinement. It should not be advertised as free or necessarily real time.

The two-route result can be used as a deployment menu. RDAR is the lighter reference. The high-performance route provides the highest mean AP and better ECE at the same parameter scale. Strict voting supplies the strongest paired no-regression evidence but incurs extra latency and poorer calibration. No single route dominates accuracy, reliability, calibration, and runtime simultaneously.

6.6. Limitations and future work

The study is limited to radar-only car detection and the frozen versions of four datasets. Other road-user classes have different dimensions and return patterns; the recovery ceiling and voting neighborhood may not transfer unchanged. The study also uses a PointPillars-style anchor detector. Testing center-based and transformer-based radar detectors would determine whether the proposal mechanisms are architecture independent.

The progressive construction analysis is based on seed 2028. It shows a coherent path but does not establish independent, monotonic, three-seed contribution for every module. A complete factorial evaluation would be expensive but would separate interactions between candidate preservation, recovery, and quality alignment more strongly. In particular, the small M2 increment warrants careful evaluation before any standalone recovery claim.

The strict route was selected from a threshold grid evaluated on the same four-dataset protocol used to report it. Neighboring all-positive settings reduce concern about a single isolated optimum, but a stronger design would select thresholds on a development set and freeze them before testing on additional held-out datasets.

Similarly, matched-seed dropout experiments for both final routes would make the stress comparison cleaner.

Future work should therefore prioritize an optimized voting kernel, held-out threshold validation, additional classes and sensors, and formal distribution-shift tests. These extensions could determine whether the observed reliability pattern generalizes. They are not implied by the current results.

7. Conclusion

This work examined proposal handling, rather than backbone scale, as a source of reliability in sparse-radar 3D object detection. The central premise was that a detector can fail at three distinct decision points: a useful candidate may disappear, its confidence may disagree with localization quality, or several locally compatible proposals may remain geometrically inconsistent. Treating these failures as proposal availability, quality, and consistency provides a clearer design target than pursuing aggregate AP with one unconstrained post-processing change.

The resulting framework assigns a restricted role to each source of evidence. M1 preserves borderline candidates, M2 recovers complementary residual boxes while capping their scores, and M3 trains confidence against a target that combines objectness with online 3D IoU. The strict endpoint then uses an expert-consistency gate and local voting that can update only the BEV center and heading of supported primary proposals. This asymmetric design is the main methodological contribution: auxiliary evidence can recover, reweight, or stabilize a detection, but it cannot silently acquire all privileges of the primary stream or enlarge the predicted geometry.

The four-dataset, three-seed evaluation supports two different conclusions. Across Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar, the strict route improves RDAR in all 12 paired comparisons. Its mean improvement is 0.9726 AP, its minimum improvement is 0.1718 AP, and the bootstrap interval for the mean paired gain is [0.6468, 1.3536]. The high-performance route raises macro AP from 35.3552 to 39.1214, a gain of 3.7662 AP, and improves ECE in the seed-2028 diagnostic audit. It nevertheless contains one negative TruckScenes seed result and is therefore not assigned the strict no-regression label.

The supporting analyses explain why the endpoints differ. The progressive seed-2028 study identifies candidate preservation and quality-aligned scoring as the main sources of average improvement, while residual recovery contributes a smaller incremental change. Calibration results favor the quality-aware routes, whereas strict voting improves paired AP without improving ECE. The closed voting grid contains multiple 12-of-12-positive configurations, so the strict result is not confined to one isolated threshold; however, stronger votes can exchange consistency for a higher macro value. Mixed range and sparsity results and one TruckScenes point-dropout failure further show that aggregate gains do not establish uniform robustness.

These findings lead to an explicit engineering choice. The high-performance route is appropriate when mean detection accuracy and confidence quality are primary and the observed seed-level regression can be validated or tolerated. The strict route is appropriate when positive paired behavior over the tested conditions is more important and an additional 22–23 ms of voting latency is acceptable. Neither route establishes universal robustness beyond the finite datasets, seeds, class, and corruption protocol studied here. Future work should freeze thresholds on an independent development set, evaluate more seeds and object classes, test additional sensors and distribution shifts, and reduce the voting cost with vectorized or spatially indexed matching. Such tests are required before the observed no-regression pattern can be promoted from bounded experimental evidence to a broader reliability guarantee.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex to assist with language editing, LaTeX formatting, and the preparation of figures from author-provided frozen experimental results. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

CRediT authorship contribution statement

Zixuan Liu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing–original draft, and Visualization. Wei Xiong: Supervision, Project administration, Funding acquisition, and Writing–review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The paper uses four radar datasets and frozen experimental outputs already recorded in the project workspace. Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar are available from their original providers subject to their respective access and license terms. Derived experimental configurations and result summaries supporting the findings are available from the corresponding author upon reasonable request.

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